About and Query+Output View

Swiggy Sales case study

This project is based on a **Swiggy Sales Case Study** dataset, which contains information related to food orders placed through the Swiggy platform. The dataset includes details such as **State, City, Order Date, Restaurant Name, Location, Category, Dish Name, Price (INR), Rating,** and **Rating Count**.

The data represents orders from different restaurants and provides insights into menu items, pricing, customer ratings, and order trends.

I have used this dataset that represents orders from different restaurants and provides insights into menu items, pricing, customer ratings, and order trends.

Data Exploration

- Viewed the complete dataset using SELECT.
- Examined the overall structure and contents of the table before starting the data cleaning process.

```
select * from Swiggy_Data
```

	State	City	Order_Date	Restaurant_Name	Location	Category	Dish_Name	Price_INR	Rating	Rating_Count
1	Karnataka	Bengaluru	2025-06-29	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butter Murukku-200gm	133.899993896484	4	0
2	Karnataka	Bengaluru	2025-04-03	Srinidhi Sagar Deluxe	Kengeri	Recommended	Badam Milk	52	4.5	25
3	Karnataka	Bengaluru	2025-01-15	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chow Chow Bath	117	4.69999980926514	48
4	Karnataka	Bengaluru	2025-04-17	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesari Bath	65	4.59999990463257	65
5	Karnataka	Bengaluru	2025-03-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Mix Raitha	130	4	0
6	Karnataka	Bengaluru	2025-07-08	Srinidhi Sagar Deluxe	Kengeri	Recommended	Srinidhi Sagar Special	312	4	0
7	Karnataka	Bengaluru	2025-01-21	Srinidhi Sagar Deluxe	Kengeri	Recommended	Garlic Naan	98	4	34
8	Karnataka	Bengaluru	2025-04-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Pista	137	4	0
9	Karnataka	Bengaluru	2025-05-02	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Butter Masala	241	4.40000009536743	29
10	Karnataka	Bengaluru	2025-07-30	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Dal Tadka	195	4.90000009536743	51
11	Karnataka	Bengaluru	2025-08-03	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Kaju Masla	312	4	0
12	Karnataka	Bengaluru	2025-03-16	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Palak Panner	234	4	0
13	Karnataka	Bengaluru	2025-07-22	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Kofta	286	4	0
14	Karnataka	Bengaluru	2025-03-29	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Burji	312	4	0
15	Karnataka	Bengaluru	2025-04-27	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Shahi Kurma	286	4.80000019073486	4
16	Karnataka	Bengaluru	2025-04-10	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Malai Kofta	260	4.5	26
17	Karnataka	Bengaluru	2025-04-14	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Navarathna Kurma (Sweet Dish)	299	4	0

Data Cleaning

- Checked for **NULL values** in all columns.

```sql
select
sum(case when State is null then 1 else 0 end) as NullStates,
sum(case when City is NULL then 1 else 0 end) as NullCity,
sum(case when Order_Date is null then 1 else 0 end) as NullOrderDates,
sum(case when Restaurant_Name is null then 1 else 0 end) as NullRestaurant_Name,
sum(case when Location is NULL then 1 else 0 end) as NullLocation,
sum(case when Category is null then 1 else 0 end) as NullCategory,
sum(case when Dish_Name is null then 1 else 0 end) as NullDish_Name,
sum(case when Price_INR is NULL then 1 else 0 end) as NullPrice_INR,
sum(case when Rating is null then 1 else 0 end) as NullRating,
sum(case when Rating_Count is null then 1 else 0 end) as NullRating_Count
from Swiggy_Data
```

100 %   No issues found

Results   Messages

	NullStates	NullCity	NullOrderDates	NullRestaurant_Name	NullLocation	NullCategory	NullDish_Name	NullPrice_INR	NullRating	NullRating_Count
1	0	0	0	0	0	0	0	0	0	0

Checked for **blank entries** in the relevant columns (excluding columns such as **Rating Count, Order Date,** and **Price**, where blank value checks were not applicable).

```sql
select * from Swiggy_Data where
State='' or City='' or Restaurant_Name='' or Location='' or
category='' or Dish_Name=''
```

State	City	Order_Date	Restaurant_Name	Location	Category	Dish_Name	Price_INR	Rating	Rating_Count

Checked the dataset for **duplicate records**.

```sql
select State,City,Order_Date, Restaurant_Name, Location, Category,Dish_Name,Price_INR,Rating, Rating_Count
, count(*) as cnt   from Swiggy_Data
group by State, City, Order_Date, Restaurant_Name, Location, Category,Dish_Name,Price_INR,Rating, Rating_Count
having count(*) >1
```

	City	Order_Date	Restaurant_Name	Location	Category	Dish_Name	Price_INR	Rating	Rating_Count	cnt
	Ahmedabad	2025-01-02	McDonald's Gourmet Burger Collection	Ghatlodia	Coffee & Beverages (Hot and Cold)	American Mud Pie Shake	205	4.40000009536743	0	2
	Ahmedabad	2025-01-16	McDonald's Gourmet Burger Collection	Ghatlodia	Burgers & Wraps	Big Spicy Paneer Wrap	241.899993896484	5	4	2
	Ahmedabad	2025-01-17	McDonald's Gourmet Burger Collection	Ghatlodia	Fries & Sides	Piri Piri Spice Mix	23.7999992370605	4.69999980926514	4	2
	Ahmedabad	2025-04-30	Veg Meals by Lunchbox	Sola	Starters and Beverages	Lemon Soda (250 mL)	49	4.40000009536743	0	2
	Ahmedabad	2025-05-11	KFC	Prahlad Nagar	Hot & Crispy Chicken & Wings	Hot & Crispy Chicken - 8 pcs	799.049987792969	3.5	79	2
	Ahmedabad	2025-06-19	Faasos' Signature Wraps & Rolls	Sola	Breakfast Specials	Double Egg Chatpata Roll	109	5	1	2
	Ahmedabad	2025-07-10	McDonald's	Ghatlodia	McSaver Combos (2 Pc Meals)	Mc Crispy Chicken Burger + Piri Piri Fries (M)	360	4	2	2
	Ahmedabad	2025-07-13	KFC	Prahlad Nagar	Recommended	Korean Tangy Chicken Roll	139	4.19999980926514	99	2
	Ahmedabad	2025-08-04	Jayhind Sweets	Ghatlodia	Namkeen	Bombay Mixer	94	4.40000009536743	0	2
	Ahmedabad	2025-08-19	McDonald's	Ghatlodia	Protein Plus and Burgers with Millet Bun	McCrispy Chicken Burger Protein Plus (2 Slices)	279	4.40000009536743	0	2
esh	Shimla	2025-06-15	Chef's Planet	Solan Town	Burgers	Cold Coffee	189	4.40000009536743	0	2
	Bengaluru	2025-01-27	The Belgian Waffle Co.	Kengeri	Recommended	Mini Waffle Box of 6 - Assorted	525	4.90000009536743	11	2
	Bengaluru	2025-02-22	Olio - The Wood Fired Pizzeria	Whitefield	Desserts	Tiramisu Doughnut Cake	139	3.90000009536743	1	2
	Bengaluru	2025-02-27	Chinese Wok	Kengeri Sate...	Drinks	7 Up	57	4.09999990463257	10	2
	Bengaluru	2025-08-13	Grameen Kulfi	KENGERI U...	Matka Kulfi	Desi Malai Matka Kulfi (pack Of 2)	152.539993286133	4.90000009536743	26	2

Deleted the duplicate records identified in the dataset.

```sql
WITH cte AS (
    SELECT *,
        ROW_NUMBER() OVER (
            PARTITION BY State, City, Order_Date, Restaurant_Name,
                         Location, Category, Dish_Name, Price_INR,
                         Rating, Rating_Count
            ORDER BY (SELECT NULL)
        ) AS rn
    FROM Swiggy_Data
)
DELETE FROM cte
WHERE rn > 1;
```

```
(0 rows affected)

Completion time: 2026-07-19T20:05:21.3433075+05:30
```

Data Normalization

To achieve proper normalization, separate dimension tables were created: Date, Location, Restaurant, Category, Dish

```sql
create table dim_date(
date_id int identity(1,1) primary key,
full_date date,
year int,
day int,
month_name varchar(100),
month int,
quarter int,
week int
)
select * from dim_date
```

	date_id	full_date	year	day	month_name	month	quarter	week

```sql
create table dim_loc(
loc_id int identity(1,1) primary key,
state varchar(50),
city varchar(50),
location varchar(50)
)
select * from dim_loc
```

Results | Messages

	loc_id	state	city	location

```sql
--created restaurent dimension table
create table restaurent_name(
restaurent_id int identity(1,1) primary key,
restaurent_name varchar(100)
)
select * from restaurent_name
```

Results | Messages

	restaurent_id	restaurent_name

```sql
--created category dimension table
create table category_name(
category_id int identity(1,1) primary key,
category_name varchar(100)
)
select * from category_name
```

Results | Messages

	category_id	category_name

```sql
--created dishname dimension table
create table dish_name(
dish_id int identity(1,1) primary key,
dish_name varchar(100)
)
select * from dish_name
```

	dish_id	dish_name

Fact Table Creation

- Created the **Fact Table**.
- Connected the Fact Table with all the Dimension Tables using **Foreign Keys** to establish proper relationships.

```sql
CREATE TABLE FACT_SWIGGY_ORDERS(
ORDER_ID INT IDENTITY(1,1)  PRIMARY KEY,

Date_id int,
Price_INR DECIMAL(10,2),
Rating decimal(4,2),
Rating_Count int,

location_id int,
restaurent_id int,
category_id int,
dish_id int,

foreign key(Date_id) references dim_date(date_id),
foreign key(location_id) references dim_loc(loc_id),
foreign key(restaurent_id) references restaurent_name(restaurent_id),
foreign key(category_id) references category_name(category_id),
foreign key(dish_id) references dish_name(dish_id)
);
select * from FACT_SWIGGY_ORDERS
```

Results | Messages

	ORDER_ID	Date_id	Price_INR	Rating	Rating_Count	location_id	restaurent_id	category_id	dish_id

Populating the Dimension Tables

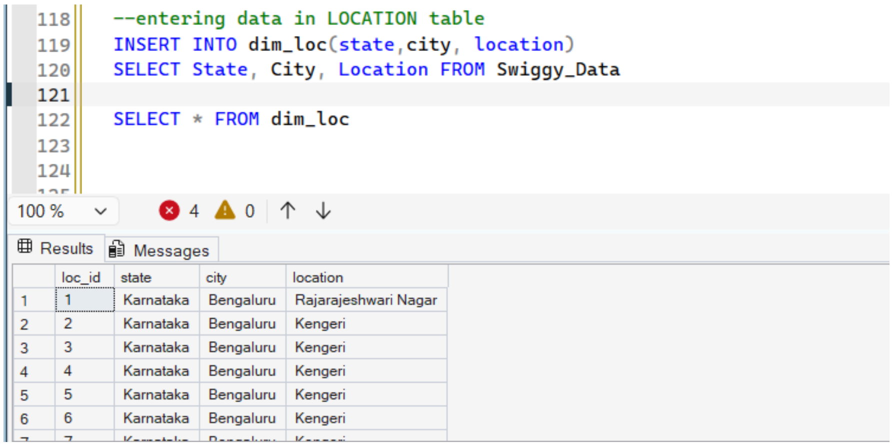
Inserted data into the following Dimension Tables: Date, Location, Restaurant, Category, Dish

```sql
--entering data in date table
insert into dim_date(full_date,year, day, month_name, MONTH,  quarter, week)
select distinct  Order_Date,
year(order_date),
day(order_date),
datename(month, Order_Date),
MONTH(ORDER_DATE),
DATEPART(QUARTER, Order_Date),
DatePart(week, Order_Date)
from Swiggy_Data
where Order_Date IS NOT NULL;

select * from dim_date
```

Results | Messages

	date_id	full_date	year	day	month_name	month	quarter	week
1	1	2025-02-18	2025	18	February	2	1	8
2	2	2025-01-05	2025	5	January	1	1	2
3	3	2025-06-02	2025	2	June	6	2	23
4	4	2025-08-17	2025	17	August	8	3	34
5	5	2025-04-23	2025	23	April	4	2	17
6	6	2025-02-15	2025	15	February	2	1	7



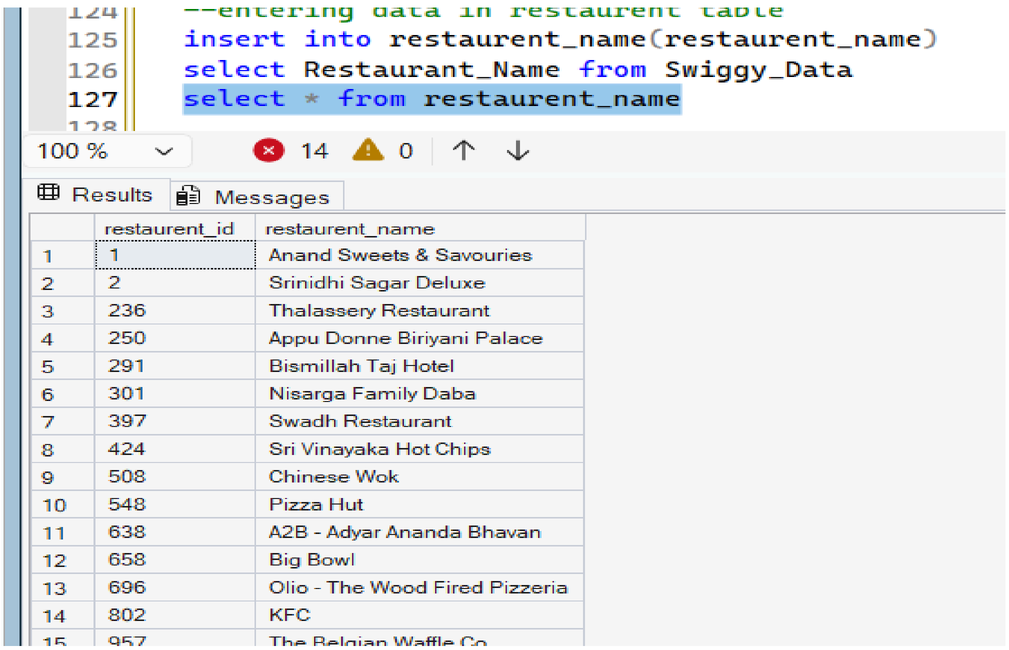
```sql
--entering data in LOCATION table
INSERT INTO dim_loc(state,city, location)
SELECT State, City, Location FROM Swiggy_Data

SELECT * FROM dim_loc
```

100 % | 4 | 0

Results | Messages

	loc_id	state	city	location
1	1	Karnataka	Bengaluru	Rajarajeshwari Nagar
2	2	Karnataka	Bengaluru	Kengeri
3	3	Karnataka	Bengaluru	Kengeri
4	4	Karnataka	Bengaluru	Kengeri
5	5	Karnataka	Bengaluru	Kengeri
6	6	Karnataka	Bengaluru	Kengeri
7	7	Karnataka	Bengaluru	Kengeri



```sql
--entering data in restaurent table
insert into restaurent_name(restaurent_name)
select Restaurant_Name from Swiggy_Data
select * from restaurent_name
```

100 % | 14 | 0

Results | Messages

	restaurent_id	restaurent_name
1	1	Anand Sweets & Savouries
2	2	Srinidhi Sagar Deluxe
3	236	Thalassery Restaurant
4	250	Appu Donne Biriyani Palace
5	291	Bismillah Taj Hotel
6	301	Nisarga Family Daba
7	397	Swadh Restaurant
8	424	Sri Vinayaka Hot Chips
9	508	Chinese Wok
10	548	Pizza Hut
11	638	A2B - Adyar Ananda Bhavan
12	658	Big Bowl
13	696	Olio - The Wood Fired Pizzeria
14	802	KFC
15	957	The Belgian Waffle Co

```sql
--entering data in CATEGORY TABLE
insert into category_name(category_name)
select DISTINCT Category from Swiggy_Data
select * from category_name
```

100 % 14 0

Results Messages

	category_id	category_name
1	1	Tea Cakes
2	2	Bakery
3	3	No Deal Like McDeal
4	4	Puffs & Bakery Bites
5	5	Veerji Special Mojito
6	6	Warps N Sandwich
7	7	Baked Cheese Pastry
8	8	Chicken Noodles
9	9	MAIN COURSE FISH
10	10	Slow Churn: Artisan...
11	11	Breads and Accom...
12	12	3 Layer Rice Bowl
13	13	Jain Special Category
14	14	No Added Sugar Mi...

```sql
--entering data in dishname TABLE
insert into dish_name(dish_name)
select DISTINCT dish_name from Swiggy_Data
select * from dish_name

ALTER TABLE Swiggy_Sales_Database.dbo.dish_name
ALTER COLUMN dish_name VARCHAR(300);
```

100 % 14 0

Results Messages

	dish_id	dish_name
1	6973	Nandhana Special Andhra Veg Carrier Meals
2	6974	Shavige Bhath + Uddina Vada(1pcs)
3	6975	Paneer Stuff Masala
4	6976	Special Dosa + 1 Vada
5	6977	Veg Khow Suey
6	6978	Egg chowmin
7	6979	Aloo Dum With Plain Paratha
8	6980	Rice Bowlz- Smoky Red Chicken
9	6981	Chicken Dum Biriyani
10	6982	Chicken Gravy
11	6983	Butter Croissant
12	6984	Mutter Paneer
13	6985	Mirchi Tipore [100 Grams]
14	6986	Cheesy fries salt and pepper
15	6987	Peppery Hot Prawns

Populating the Fact Table

- Inserted all the required data into the **Fact Table** by referencing the corresponding Dimension Tables.

```sql
--inserting data into fact table
insert into FACT_SWIGGY_ORDERS(
Date_id,
Price_INR,
Rating,
Rating_Count,
location_id,
restaurent_id,
category_id,
dish_id
)

select dd.date_id,
s.price_INR,
s.Rating,
s.Rating_Count,

d1.loc_id,
dr.restaurent_id,
dc.category_id,
dsh.dish_id
from Swiggy_Data s
join dim_date dd on dd.full_date= s.Order_Date

join dim_loc d1 on d1.location=s.Location
```

```sql
join dim_loc d1 on d1.location=s.Location
and d1.state=s.state
and d1.city=s.City

join restaurent_name dr on
dr.restaurent_name=s.Restaurant_Name

join category_name  dc on
dc.category_name =s.Category

join dish_name dsh on
dsh.dish_name=s.Dish_Name


select * from FACT_SWIGGY_ORDERS
```

110 % 14 0

Results Messages

	ORDER_ID	Date_id	Price_INR	Rating	Rating_Count	location_id	restaurent_id	category_id	dish_id
1	124596138	195	133.90	4.00	0	1	1	1338	37431
2	124596139	195	133.90	4.00	0	236	1	1338	37431
3	124596140	195	133.90	4.00	0	237	1	1338	37431
4	124596141	195	133.90	4.00	0	238	1	1338	37431
5	124596142	195	133.90	4.00	0	239	1	1338	37431
6	124596143	195	133.90	4.00	0	240	1	1338	37431
7	124596144	195	133.90	4.00	0	10451	1	1338	37431
8	124596145	195	133.90	4.00	0	10452	1	1338	37431
9	124596146	195	133.90	4.00	0	10453	1	1338	37431
10	124596147	195	133.90	4.00	0	10454	1	1338	37431
11	124596148	195	133.90	4.00	0	10455	1	1338	37431
12	124596149	195	133.90	4.00	0	10456	1	1338	37431
13	124596150	195	133.90	4.00	0	10445	1	1338	37431
14	124596151	195	133.90	4.00	0	10446	1	1338	37431
15	124596152	195	133.90	4.00	0	10447	1	1338	37431
16	124596153	195	133.90	4.00	0	10448	1	1338	37431
17	124596154	195	133.90	4.00	0	10449	1	1338	37431
18	124596155	195	133.90	4.00	0	10450	1	1338	37431
19	124596156	195	133.90	4.00	0	10439	1	1338	37431
20	124596157	195	133.90	4.00	0	10440	1	1338	37431
21	124596158	195	133.90	4.00	0	10441	1	1338	37431
22	124596159	195	133.90	4.00	0	10442	1	1338	37431
23	124596160	195	133.90	4.00	0	10443	1	1338	37431
24	124596161	195	133.90	4.00	0	10444	1	1338	37431
25	124596162	195	133.90	4.00	0	10433	1	1338	37431
26	124596163	195	133.90	4.00	0	10434	1	1338	37431
27	124596164	195	133.90	4.00	0	10435	1	1338	37431
28	124596165	195	133.90	4.00	0	10436	1	1338	37431
29	124596166	195	133.90	4.00	0	10437	1	1338	37431

Data Verification

- Viewed the complete **Fact Table** to verify that all records were inserted correctly and that the relationships with the Dimension Tables were established successfully.

```sql
select * from FACT_SWIGGY_ORDERS f
join dim_date d on f.Date_id=d.date_id
join dim_loc l on f.location_id=l.loc_id
join restaurent_name r on f.restaurent_id=r.restaurent_id
join category_name c on f.category_id=c.category_id
join dish_name dsh on f.dish_id=dsh.dish_id;
```

Results Messages

	ORDER_ID	Date_id	Price_INR	Rating	Rating_Count	location_id	restaurent_id	category_id	dish_id	date_id	full_date	year
1	124596138	195	133.90	4.00	0	1	1	1338	37431	195	2025-06-29	2025
2	124596139	195	133.90	4.00	0	236	1	1338	37431	195	2025-06-29	2025
3	124596140	195	133.90	4.00	0	237	1	1338	37431	195	2025-06-29	2025
4	124596141	195	133.90	4.00	0	238	1	1338	37431	195	2025-06-29	2025
5	124596142	195	133.90	4.00	0	239	1	1338	37431	195	2025-06-29	2025
6	124596143	195	133.90	4.00	0	240	1	1338	37431	195	2025-06-29	2025
7	124596144	195	133.90	4.00	0	10451	1	1338	37431	195	2025-06-29	2025
8	124596145	195	133.90	4.00	0	10452	1	1338	37431	195	2025-06-29	2025
9	124596146	195	133.90	4.00	0	10453	1	1338	37431	195	2025-06-29	2025
10	124596147	195	133.90	4.00	0	10454	1	1338	37431	195	2025-06-29	2025

	day	month_name	month	quarter	week	loc_id	state	city	location	restaurent_id	restaurent_name	category_id	category_r
1	29	June	6	2	27	1	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
2	29	June	6	2	27	236	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
3	29	June	6	2	27	237	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
4	29	June	6	2	27	238	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
5	29	June	6	2	27	239	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
6	29	June	6	2	27	240	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
7	29	June	6	2	27	10451	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
8	29	June	6	2	27	10452	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
9	29	June	6	2	27	10453	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack
10	29	June	6	2	27	10454	Karnataka	Bengaluru	Rajarajeshwari Nagar	1	Anand Sweets & Savouries	1338	Snack

Business Problem Analysis

After preparing and normalizing the dataset, the following business problems were analyzed using SQL.

Key Performance Indicators (KPIs)

- Find the **Total Orders**.

```sql
--KPI
--Total no. of orders
select count(*) as total_orders from FACT_SWIGGY_ORDERS
```

	total_orders
1	124596137

- Find the **Total Revenue**.

```sql
-- Total Revenue
select FORMAT(sum(convert(float, Price_INR))/100000,'N2')+'INR Million' as Total_Revenue
from FACT_SWIGGY_ORDERS
```

	Total_Revenue
1	337,211.51INR Million

- Calculate the **Average Revenue**.

```sql
--avg dish price
select format(avg(convert(float, Price_INR))/100000,'N2')+'INR Million' as Average_Dish_Price
from fact_swiggy_orders
```

	Average_Dish_PriceS
1	270.64INR Million

- Calculate the **Average Rating**.

```
--AVG RATINGS
select avg(Rating) as avg_rating from FACT_SWIGGY_ORDERS
```

	avg_rating
1	4.341599

Deep Dive Analysis

Time-Based Analysis

- Monthly Total Order Trend.

```
--Monthly trend for total orders
select dd.year, dd.month, dd.month_name, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_date dd on f.Date_id=dd.date_id
group by
dd.year, dd.month, dd.month_name
order by Total_orders desc
```

Results | Messages

	year	month	month_name	Total_orders
1	2025	1	January	16134735
2	2025	5	May	15901307
3	2025	8	August	15895145
4	2025	7	July	15632451
5	2025	4	April	15445477
6	2025	6	June	15394290
7	2025	2	February	15119710
8	2025	3	March	15073022

- Monthly Total Revenue Trend.

```sql
--Monthly trend for total revenue
select dd.year, dd.month, dd.month_name, sum(price_INR) as Total_REVENUE from FACT_SWIGGY_ORDERS f
join dim_date dd on f.Date_id=dd.date_id
group by
dd.year, dd.month, dd.month_name
order by Total_REVENUE desc
```

	year	month	month_name	Total_REVENUE
1	2025	1	January	4358754428.23
2	2025	5	May	4315268750.29
3	2025	8	August	4275289706.47
4	2025	7	July	4206479294.99
5	2025	6	June	4185839564.67
6	2025	4	April	4170285098.31
7	2025	3	March	4114101405.90
8	2025	2	February	4095132319.02

- Quarterly Total Order Trend.

```sql
--Order analysis by quarters
select dd.year, dd.quarter, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_date dd on f.Date_id=dd.date_id
group by
dd.year, dd.quarter
order by Total_orders desc
```

	year	quarter	Total_orders
1	2025	2	46741074
2	2025	1	46327467
3	2025	3	31527596

- Yearly Total Order Trend.

```sql
--order analysis by year
select dd.year, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_date dd on f.Date_id=dd.date_id
group by
dd.year
order by Total_orders desc
```

Results | Messages

	year	Total_orders
1	2025	124596137

- Weekly Order Trend.

```sql
--order analysis by week
select datename(WEEKDAY,dd.full_date) as day_name,
count(*) as total_orders from fact_swiggy_orders f
join dim_date dd on dd.date_id= f.Date_id
group by datename(WEEKDAY,dd.full_date)
order by count(*) desc
```

	day_name	total_orders
1	Saturday	18517195
2	Thursday	17950868
3	Sunday	17817106
4	Friday	17747682
5	Wednesday	17739566
6	Tuesday	17414781
7	Monday	17408939

Location-Based Analysis

- Top Cities by Total Orders.

```sql
--Top 10 location by orders
select l.city,  count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_loc l on l.loc_id= f.location_id
group by city
order by count(*) desc
```

Results | Messages

	city	Total_orders
1	Bengaluru	17774646
2	Mumbai	10337089
3	Lucknow	8912124
4	Jaipur	8794763
5	Kolkata	7349918
6	Ahmedabad	7185901
7	Chennai	5367148
8	New Delhi	5280597
9	Shimla	5167115
10	Hyderabad	4733744
11	Shillong	4604887
12	Gurgaon	3873371
13	Chandigarh	3715166
14	Indore	3656183
15	Patna	3379088
16	Ranchi	3228017
17	Raipur	3138273

- Order Analysis by State.

```sql
--Top 10 states by orders
select l.State,  count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_loc l on l.loc_id= f.location_id
group by state
order by count(*) desc
```

	State	Total_orders
1	Karnataka	17774646
2	Maharashtra	10337089
3	Uttar Pradesh	8912124
4	Rajasthan	8794763
5	West Bengal	7349918
6	Gujarat	7185901
7	Tamil Nadu	5367148
8	Delhi	5280597
9	Himachal Pradesh	5167115
10	Telangana	4733744
11	Meghalaya	4604887
12	Haryana	3873371
13	Punjab	3715166
14	Madhya Pradesh	3656183
15	Bihar	3379088
16	Jharkhand	3228017

Restaurant & Product Analysis

- Top Restaurants by Total Orders.

```
--Top restaurents by orders
select r.restaurent_name, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join restaurent_name r on r.restaurent_id=f.restaurent_id
group by restaurent_name
order by count(*) desc
```

Results | Messages

	restaurent_name	Total_orders
1	McDonald's	9892027
2	KFC	7216126
3	LunchBox - Meals and Thalis	4022842
4	Burger King	3894245
5	Pizza Hut	3563775
6	Faasos - Wraps, Rolls & Shawarma	2791308
7	Domino's Pizza	2696131
8	The Good Bowl	2120763
9	Veg Meals by Lunchbox	1741879
10	Olio - The Wood Fired Pizzeria	1613686
11	Sweet Truth - Cake and Desserts	1569887
12	Baskin Robbins - Ice Cream Desserts	1514529
13	BOX8 - Desi Meals	1138884
14	FreshMenu	1092751
15	Shiv Sagar Veg Restaurant	1034264
16	Faasos' Signature Wraps & Rolls	928731
17	Burger Farm	914831

- Top Food Categories by Total Orders.

```
--Top food_categories by orders
select c.category_name, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join category_name c on c.category_id=f.category_id
group by category_name
order by count(*) desc
```

Results | Messages

	category_name	Total_order
1	Recommended	14355979
2	Desserts	2446556
3	Main Course	1897850
4	McSaver Combos (2 Pc Meals)	1574441
5	Beverages	1563771
6	BURGERS	1417166
7	Sweets	1142197
8	Burger Combos (3 Pc Meals)	1119558
9	Starters	1063146
10	Coffee & Beverages (Hot and Cold)	1029951
11	Exclusive Deals (Save upto 40%)	942303
12	Burgers & Wraps	915805
13	ROLLS	880700
14	Breads	869873
15	Protein Plus and Burgers with Millet Bun	818767
16	DESSERTS & BEVERAGES	763212
17	Snacks	738589

Query executed successfully.

- Most Ordered Dishes.

```
--Most ordered dishes
select dsh.dish_name, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join  dish_name dsh on dsh.dish_id=f.dish_id
group by dish_name
order by count(*) desc
```

- Cuisine Performance Analysis (Total Orders and Average Ratings).

```
--Cuisine performance
select dsh.dish_name, count(*) as Total_orders, avg(rating) As avg_rating from FACT_SWIGGY_ORDERS f
join dish_name dsh on dsh.disch_id=f.dish_id
group by dish_name
order by count(*) desc
```

Results | Messages

	dish_name	Total_orders	avg_rating
1	Choco Lava Cake	201412	4.387960
2	Veg Fried Rice	175414	4.247102
3	Jeera Rice	161556	4.413425
4	Paneer Butter Masala	159441	4.135201
5	French Fries	143000	4.229653
6	Butter Naan	125678	4.255267
7	Cold Coffee	124959	4.254424
8	Green Salad	116757	4.420209
9	Chicken Sausage	115873	4.326337
10	Mango Cheesecake	112418	4.416232
11	Chicken Fried Rice	108965	4.334076
12	Dal Makhani	108852	4.296204
13	Walnut Brownie	103320	4.441985
14	Choco Chip Brownie	102930	4.402273
15	Margherita Pizza	102922	4.221102
16	Blueberry Cheesecake	99291	4.382114
17	Double Chicken Roll	97998	4.290991

Query executed successfully.

Price & Rating Analysis

- Number of Orders Across Different Price Ranges.

```sql
--order count by price range
select
case
when convert(float,PRICE_INR)<100 THEN 'Less than 100'
 when convert(float, price_INR)  BETWEEN 100 AND 200 THEN ' 100-199'
 WHEN CONVERT(FLOAT, PRICE_INR)  BETWEEN 200 AND 300 THEN '200-299'
 WHEN CONVERT( FLOAT, PRICE_INR)  BETWEEN 300 AND 400 THEN '300-399'
 WHEN CONVERT( FLOAT, PRICE_INR)  BETWEEN 400 AND 500 THEN '400-499'
ELSE '500 AND ABOVE'
END AS PRICE_RANGE, COUNT(*) AS TOTAL_ORDERS
FROM FACT_SWIGGY_ORDERS
GROUP BY
case
 when convert(float,Price_INR)<100 THEN 'Less than 100'
 when convert(float, price_INR)  BETWEEN 100 AND 200 THEN ' 100-199'
 WHEN CONVERT(FLOAT, PRICE_INR)  BETWEEN 200 AND 300 THEN '200-299'
 WHEN CONVERT( FLOAT, PRICE_INR)  BETWEEN 300 AND 400 THEN '300-399'
 WHEN CONVERT( FLOAT, PRICE_INR)  BETWEEN 400 AND 500 THEN '400-499'
ELSE '500 AND ABOVE'
END
ORDER BY COUNT(*) DESC
```

Results | Messages

	PRICE_RANGE	TOTAL_ORDERS
1	100-199	36059624
2	200-299	35696229
3	300-399	19530369
4	Less than 100	16022673
5	500 AND ABOVE	9718990
6	400-499	7568252

Query executed successfully.

- Order Count for Each Rating.

```
--order counts per ratings
select Rating,
count(*) as Total_orders from   FACT_SWIGGY_ORDERS
group by Rating
order by Rating asc
```

Results | Messages

	Rating	Total_orders
1	1.50	44481
2	1.60	29469
3	1.70	17459
4	1.80	42407
5	1.90	38222
6	2.00	161703
7	2.10	109746
8	2.20	124630

Query executed successfully.

SQL Code

```sql
select * from Swiggy_Data

select
sum(case when State is null then 1 else 0 end) as NullStates,
sum(case when City is NULL then 1 else 0 end) as NullCity,
sum(case when Order_Date is null then 1 else 0 end) as NullOrderDates,
sum(case when Restaurant_Name is null then 1 else 0 end) as NullRestaurant_Name,
sum(case when Location is NULL then 1 else 0 end) as NullLocation,
sum(case when Category is null then 1 else 0 end) as NullCategory,
sum(case when Dish_Name is null then 1 else 0 end) as NullDish_Name,
sum(case when Price_INR is NULL then 1 else 0 end) as NullPrice_INR,
sum(case when Rating is null then 1 else 0 end) as NullRating,
sum(case when Rating_Count is null then 1 else 0 end) as NullRating_Count
from Swiggy_Data

select * from Swiggy_Data where
State='' or City='' or Restaurant_Name='' or Location='' or
category='' or Dish_Name=''
select State,City,Order_Date, Restaurant_Name, Location, Category,Dish_Name,Price_INR,Rating, Rating_Count
, count(*) as cnt   from Swiggy_Data
group by State, City, Order_Date, Restaurant_Name, Location, Category,Dish_Name,Price_INR,Rating,
Rating_Count
having count(*) >1


WITH cte AS (
   SELECT *,
      ROW_NUMBER() OVER (
         PARTITION BY State, City, Order_Date, Restaurant_Name,
                     Location, Category, Dish_Name, Price_INR,
                     Rating, Rating_Count
         ORDER BY (SELECT NULL)
      ) AS rn
   FROM Swiggy_Data
)
DELETE FROM cte
WHERE rn > 1;

create table dim_date(
date_id int identity(1,1) primary key,
full_date date,
year int,
day int,
month_name varchar(100),
month int,
quarter int,
week int
)
select * from dim_date

create table dim_loc(
loc_id int identity(1,1) primary key,
state varchar(50),
city varchar(50),
```

```
location varchar(50)
)
select * from dim_loc

--created restaurent dimension table
create table restaurent_name(
restaurent_id int identity(1,1) primary key,
restaurent_name varchar(100)
)
select * from restaurent_name

--created category dimension table
create table category_name(
category_id int identity(1,1) primary key,
category_name varchar(100)
)
select * from category_name

--created dishname dimension table
create table dish_name(
dish_id int identity(1,1) primary key,
dish_name varchar(100)
)
select * from dish_name

--CREATING FACT TABLE

select * from Swiggy_Data

CREATE TABLE FACT_SWIGGY_ORDERS(
ORDER_ID INT IDENTITY(1,1)  PRIMARY KEY,

Date_id int,
Price_INR DECIMAL(10,2),
Rating decimal(4,2),
Rating_Count int,

location_id int,
restaurent_id int,
category_id int,
dish_id int,

foreign key(Date_id) references dim_date(date_id),
foreign key(location_id) references dim_loc(loc_id),
foreign key(restaurent_id) references restaurent_name(restaurent_id),
foreign key(category_id) references category_name(category_id),
foreign key(dish_id) references dish_name(dish_id)
);
select * from FACT_SWIGGY_ORDERS

--entering data in date table
insert into dim_date(full_date,year, day, month_name, MONTH,  quarter, week)
select distinct  Order_Date,
year(order_date),
```

```
day(order_date),
datename(month, Order_Date),
MONTH(ORDER_DATE),
DATEPART(QUARTER, Order_Date),
DatePart(week, Order_Date)
from Swiggy_Data
where Order_Date IS NOT NULL;

select * from dim_date

--entering data in LOCATION table
INSERT INTO dim_loc(state,city, location)
SELECT State, City, Location FROM Swiggy_Data

SELECT * FROM dim_loc

--entering data in restaurent table
insert into restaurent_name(restaurent_name)
select Restaurant_Name from Swiggy_Data
select * from restaurent_name

WITH R_Duplicates AS (
   SELECT restaurent_name,
         ROW_NUMBER() OVER (
            PARTITION BY restaurent_namE
            ORDER BY restaurent_id ASC
         ) AS row_num
   FROM restaurent_name
)
DELETE FROM R_Duplicates
WHERE row_num > 1;

--entering data in CATEGORY TABLE
insert into category_name(category_name)
select DISTINCT Category from Swiggy_Data
select * from category_name

--entering data in dishname TABLE
insert into dish_name(dish_name)
select DISTINCT dish_name from Swiggy_Data
select * from dish_name

ALTER TABLE Swiggy_Sales_Database.dbo.dish_name
ALTER COLUMN dish_name VARCHAR(300);

--inserting data into fact table
insert into FACT_SWIGGY_ORDERS(
Date_id,
Price_INR,
Rating,
Rating_Count,
location_id,
restaurent_id,
category_id,
```

```sql
dish_id
)

select dd.date_id,
s.price_INR,
s.Rating,
s.Rating_Count,

d1.loc_id,
dr.restaurent_id,
dc.category_id,
dsh.dish_id
from Swiggy_Data s
join dim_date dd on dd.full_date= s.Order_Date

join dim_loc d1 on d1.location=s.Location
and d1.state=s.state
and d1.city=s.City

join restaurent_name dr on
dr.restaurent_name=s.Restaurant_Name

join category_name  dc on
dc.category_name =s.Category

join dish_name dsh on
dsh.dish_name=s.Dish_Name


select * from FACT_SWIGGY_ORDERS
--viewing full fact table

select * from FACT_SWIGGY_ORDERS f
join dim_date d on f.Date_id=d.date_id
join dim_loc l on f.location_id=l.loc_id
join restaurent_name r on f.restaurent_id=r.restaurent_id
join category_name c on f.category_id=c.category_id
join dish_name dsh on f.dish_id=dsh.dish_id;

--KPI
--Total no. of orders
select count(*) as total_orders from FACT_SWIGGY_ORDERS

-- Total Revenue
select FORMAT(sum(convert(float, Price_INR))/100000,'N2')+'INR Million' as Total_Revenue
from FACT_SWIGGY_ORDERS

--avg dish price
select format(avg(convert(float, Price_INR)),'N2')+'INR Million' as Average_Dish_PriceS
from fact_swiggy_orders

--AVG RATINGS
select avg(Rating) as avg_rating from FACT_SWIGGY_ORDERS
```

```sql
--Monthly trend for total revenue
select dd.year, dd.month, dd.month_name, sum(price_INR) as Total_REVENUE from FACT_SWIGGY_ORDERS f
join dim_date dd on f.Date_id=dd.date_id
group by
dd.year, dd.month, dd.month_name
order by Total_REVENUE desc

--Order analysis by quarters
select dd.year, dd.quarter, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_date dd on f.Date_id=dd.date_id
group by
dd.year, dd.quarter
order by Total_orders desc

--order analysis by year
select dd.year, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_date dd on f.Date_id=dd.date_id
group by
dd.year
order by Total_orders desc

--order analysis by week
select datename(WEEKDAY,dd.full_date) as day_name,
count(*) as total_orders from fact_swiggy_orders f
join dim_date dd on dd.date_id= f.Date_id
group by datename(WEEKDAY,dd.full_date)
order by count(*) desc

--Top  location by orders
select l.city,  count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_loc l on l.loc_id= f.location_id
group by city
order by count(*) desc

--Top  states by orders
select l.State,  count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join dim_loc l on l.loc_id= f.location_id
group by state
order by count(*) desc

--Top restaurents by orders
select r.restaurent_name, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join restaurent_name r on r.restaurent_id=f.restaurent_id
group by restaurent_name
order by count(*) desc

--Top food_categories by orders
select c.category_name, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
join category_name c on c.category_id=f.category_id
group by category_name
order by count(*) desc

--Most ordered dishes
select dsh.dish_name, count(*) as Total_orders from FACT_SWIGGY_ORDERS f
```

```sql
join  dish_name dsh on dsh.dish_id=f.dish_id
group by dish_name
order by count(*) desc

--Cuisine performance
select dsh.dish_name, count(*) as Total_orders, avg(rating) As avg_rating from FACT_SWIGGY_ORDERS f
join dish_name dsh on dsh.dish_id=f.dish_id
group by dish_name
order by count(*) desc

--order count by price range
select
case
when convert(float,PRICE_INR)<100 THEN 'Less than 100'
 when convert(float, price_INR)  BETWEEN 100 AND 200 THEN ' 100-199'
 WHEN CONVERT(FLOAT, PRICE_INR)  BETWEEN 200 AND 300 THEN '200-299'
 WHEN CONVERT( FLOAT, PRICE_INR)  BETWEEN 300 AND 400 THEN '300-399'
 WHEN CONVERT( FLOAT, PRICE_INR)  BETWEEN 400 AND 500 THEN '400-499'
ELSE '500 AND ABOVE'
END AS PRICE_RANGE, COUNT(*) AS TOTAL_ORDERS
FROM FACT_SWIGGY_ORDERS
GROUP BY
case
 when convert(float,Price_INR)<100 THEN 'Less than 100'
 when convert(float, price_INR)  BETWEEN 100 AND 200 THEN ' 100-199'
 WHEN CONVERT(FLOAT, PRICE_INR)  BETWEEN 200 AND 300 THEN '200-299'
 WHEN CONVERT( FLOAT, PRICE_INR)  BETWEEN 300 AND 400 THEN '300-399'
 WHEN CONVERT( FLOAT, PRICE_INR)  BETWEEN 400 AND 500 THEN '400-499'
ELSE '500 AND ABOVE'
END
ORDER BY COUNT(*) DESC

--order counts per ratings
select Rating,
count(*) as Total_orders from   FACT_SWIGGY_ORDERS
group by Rating
order by Rating asc
```